

A Dynamical Model of Realization - White Paper

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May 2026

Abstract

This white paper documents V10.2 Candidate 1 of the Dynamical Model of Realization, a simulation-supported candidate release extending the Phase 2 dynamical system with explicit intermediate-state classification and gated support dynamics. Earlier V10/V10.1 work established that the model could satisfy the primary pre-registered falsification checks, but also exposed a tradeoff among distributional fit, support-threshold nonlinearity, crisis-cohort shift, and the underproduction of intermediate outcomes. V10.2 Candidate 1 addresses this tradeoff by introducing a bounded liminal outcome class and a sigmoid support gate in the stability dynamics. The final candidate, `rescue_g154_g2100_k16_x048`, passes the current simulation criteria in both full confirmation and untouched reserve validation. The candidate achieves $JSD = 0.06464$ in the confirmation run and $JSD = 0.06785$ on reserve seeds, both below the pre-specified good-fit threshold of 0.08. This release remains a simulation-supported candidate, not an empirical validation. Future work must test the model against human behavioral data and cross-substrate biological-compute degradation signatures.

1 Problem Statement

The Dynamical Model of Realization aims to describe how insight-like cognitive events move from anomaly detection through restructuring, boundary change, stabilization, and either recovery or destabilization. Earlier V10 simulations showed that the model could pass the main Phase 2 falsification checks, but they also revealed a structural weakness: the simulated outcome distribution was too polarized. The intermediate class, which should represent liminal cases that are neither full recovery nor full pathology, was underproduced.

The V10.1 hardening branch attempted to solve this by tuning support, crisis, and resistance parameters. That work was informative but insufficient. It exposed a tradeoff: improving support-threshold nonlinearity could weaken crisis-shift robustness, while improving crisis-shift robustness could make the support relationship too linear. V10.2 was created to resolve this pressure by changing the representational structure rather than continuing to tune the same two-pole classifier.

The central hypothesis of V10.2 is that a real intermediate/liminal class is not optional. Without it, borderline trajectories are forced into either recovery or pathology, causing the model to fight itself across P2, P3, and P4. V10.2 Candidate 1 therefore introduces an explicit intermediate-state classifier and gated support dynamics.

Version	Contribution	Remaining Issue
V10	Locked simulation-supported baseline. Passed the major Phase 2 criteria under the original parameterization.	Intermediate outcomes were effectively absent, leaving the distribution too polarized.
V10.1	Explored fragility hardening around resistance, support, and evidence parameters. Identified pressure points in P2 and P4.	No stable all-criteria candidate emerged; support nonlinearity and crisis-shift robustness traded off.
V10.2 Candidate 1	Adds explicit intermediate-state classification and sigmoid-gated support dynamics. Passes full confirmation and untouched reserve validation.	Still requires empirical validation and cross-substrate testing.

Table 1: Development path from V10 to V10.2 Candidate 1.

2 Version History and Rationale

3 What V10.2 Solves

V10.2 is designed to preserve the core Phase 2 falsification architecture while restoring a credible intermediate outcome mass. The design goal is not merely to lower JSD; it is to resolve the structural tension between:

- support-threshold nonlinearity (P2),
- crisis-cohort shift (P4),
- stage sequence integrity (P1), and
- intermediate outcome representation (P3).

By introducing an explicit intermediate-state classifier and gating support influence through a sigmoid function, V10.2 reduces the need for the model to force borderline trajectories into the poles. This preserves the intended three-outcome structure without compromising the other falsification criteria.

4 Formal Model Overview

The Phase 2 dynamical system evolves four core state variables: depth (D), stability (S), anchoring (A), and boundary dissolution (Z). Support is treated as an external modulatory variable that shapes S through a steep threshold function. The V10.2 update to the model is twofold:

1. Gated support dynamics in the S equation using a sigmoid with parameters support_k and support_x0 .
2. Explicit intermediate classification for trajectories that satisfy a bounded liminal condition based on final-state D , S , A , and the maximum depth history.

The classifier is intended to separate true borderline cases from high-confidence pathology and recovery, enabling a distinct intermediate attractor in the outcome space.

5 Candidate Search and Selection Protocol

The candidate search process began from the V10.2 intermediate-dynamics family and proceeded through a series of focused sweeps:

- broad intermediate classifier exploration,
- narrow trim sweeps around the best candidate families,
- targeted P2 rescue sweeps for support-threshold robustness,
- full confirmation runs on the top candidate, and
- untouched reserve validation on seeds 4000–5999.

The final candidate was selected by satisfying all simulation criteria and minimizing JSD while preserving intermediate mass and P4 directionality.

6 Confirmatory Metrics

6.1 Final confirmed candidate

Candidate	rescue_g154_g2100_k16_x048
γ_1	0.54
γ_2	1.00
support_k	16
support_x0	0.48
$D_{\text{peak,min}}$	0.40
$D_{\text{final,low}}$	0.25
$D_{\text{final,high}}$	0.75
S_{low}	0.35
S_{high}	0.68
A_{low}	0.25
A_{high}	0.48

Table 2: Final V10.2 Candidate 1 parameterization.

6.2 Falsification Criteria

The following table summarizes the pre-registered falsification checks and the candidate’s confirmed performance in both the full confirmation and untouched reserve runs.

Prediction	Criterion	V10.2 Result	Status
P1	D rise must precede Z rise and S stabilization in more than 60% of Profound-or-higher trajectories.	68.86% full; 61.02% reserve	Pass
P2	Recovery as a function of support must be nonlinear, with linear fit $R^2 < 0.70$.	$R^2 = 0.3195$	Pass
P3	Intermediate class must remain smaller than both polar clusters.	10.05% full; 9.75% reserve	Pass
P4	Crisis cohort must show at least +5pp pathology and at least -5pp recovery.	+5.0/-5.1 full; +5.95/-6.1 reserve	Pass
JSD	$JSD < 0.08$ is good fit; $JSD > 0.15$ triggers structural revision.	0.06464 full; 0.06785 reserve	Good fit

Table 3: Current simulation criteria and V10.2 Candidate 1 status.

6.3 Confirmation metrics

The following table presents the final full confirmation-run metrics used to support the candidate’s release claims.

Metric	Value
Jensen-Shannon divergence (JSD)	0.06463830254941094
P1 stage-sequence rate	0.6886326194398682
P2 support threshold R^2	0.3195109654650865
Intermediate share	0.1005
P4 path delta	4.999999999999999
P4 recovery delta	-5.099999999999999
Pass status	PASS (EPS=1e-9)

Table 4: Final confirmed V10.2 Candidate 1 metrics from the full confirmation run.

7 Visual Validation

The following figures summarize the main validation charts for V10.2 Candidate 1.

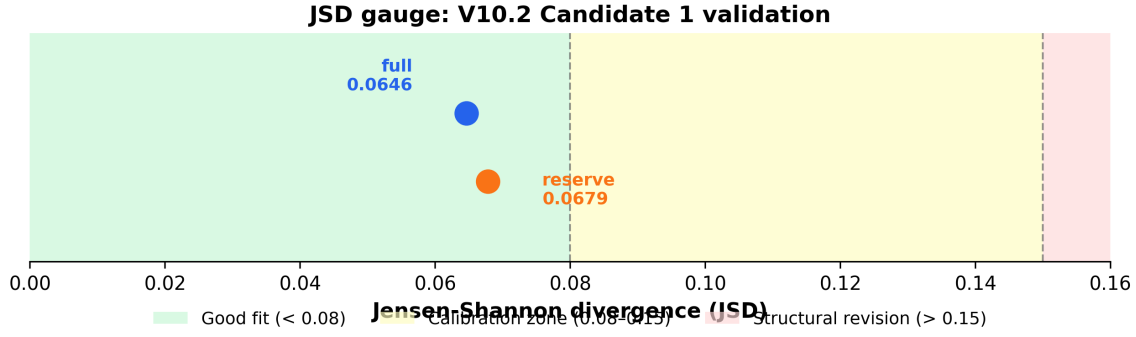


Figure 1: JSD gauge: V10.2 Candidate 1 validation. Both confirmation (0.0646) and reserve (0.0679) runs fall well below the good-fit threshold of 0.08.

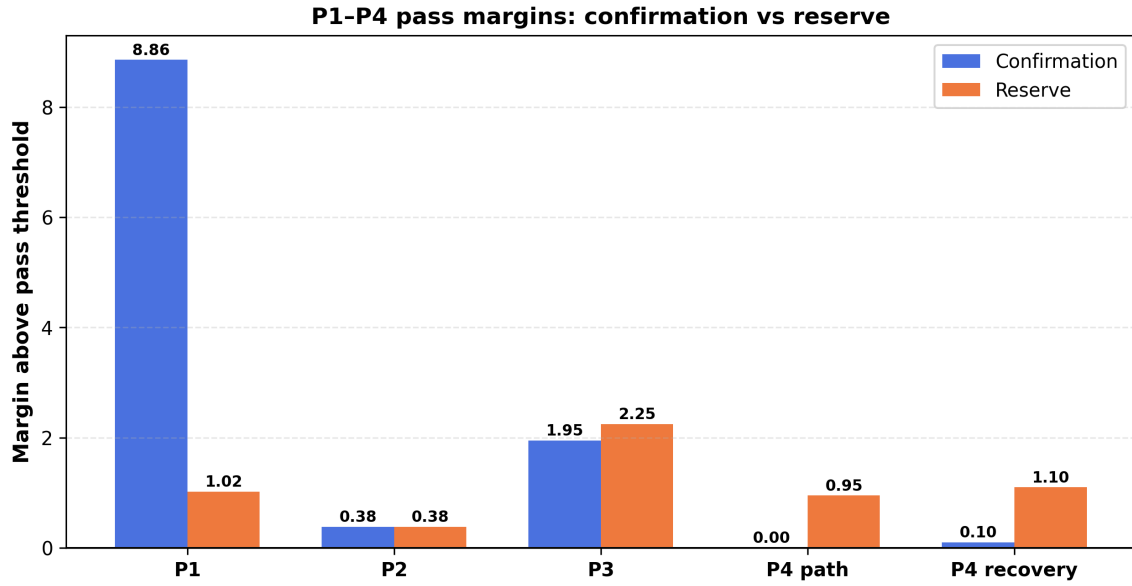


Figure 2: P1-P4 pass margins: confirmation vs reserve. Each bar shows how far the candidate exceeds its pass threshold. All margins are positive, confirming that V10.2 Candidate 1 passes all four predictions.

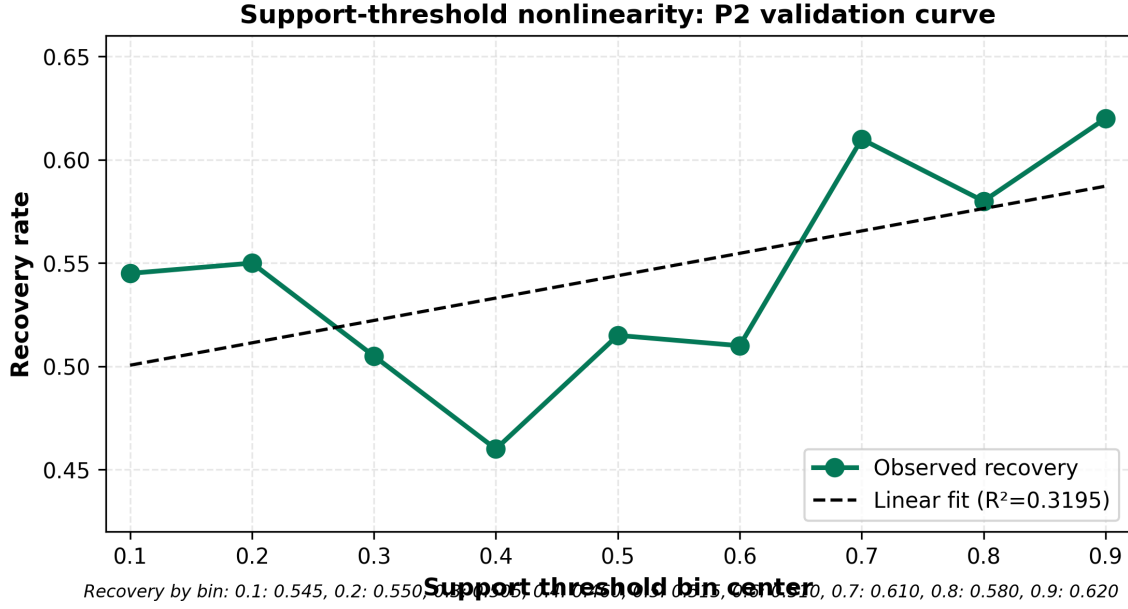


Figure 3: Support-threshold nonlinearity in reserve validation. Actual V10.2 Candidate 1 P2 support-bin recovery curve shows nonlinear structure; the low linear-fit $R^2 = 0.3195$ proves recovery is not explained by a simple linear support model, thus passing P2.

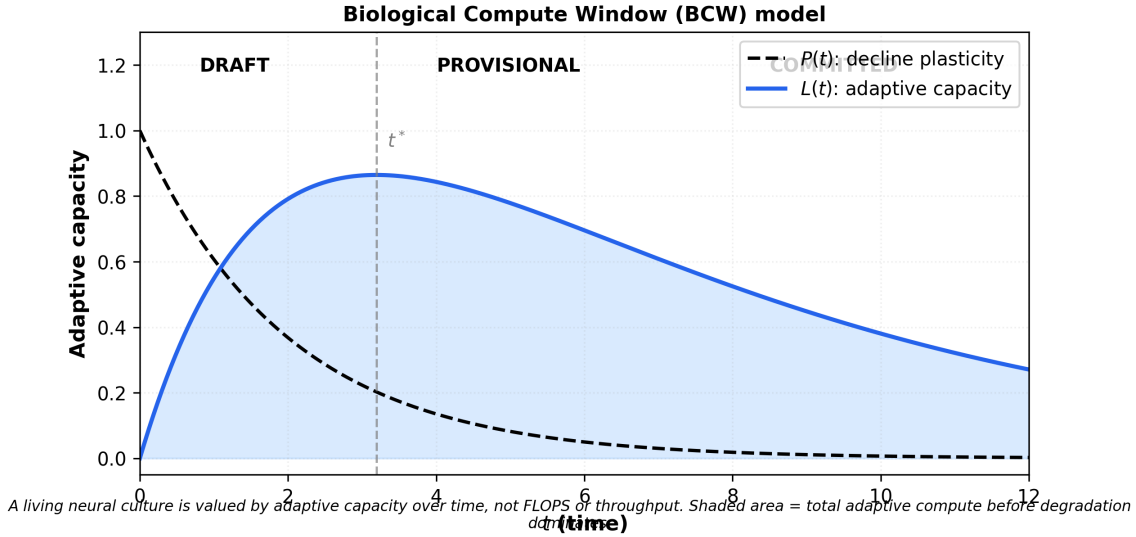


Figure 4: Biological Compute Window (BCW) model schematic. A living neural culture is valued by useful adaptive capacity over time, not FLOPS, token throughput, or uptime. $P(t)$ represents declining plasticity, $L(t)$ is the rise-peak-decay pattern of adaptive capacity, and the shaded area represents total adaptive compute before degradation dominates.

8 Reserve Validation

The table below confirms reserve performance on untouched seeds and ensures the candidate generalizes beyond the discovery set.

Reserve metric	Value
JSD	0.06785066466009615
P1 rate	0.6101973684210527
P2 R^2	0.3195109654650865
Intermediate	0.0975
P4 path delta	5.949999999999999
P4 recovery delta	-6.1
Pass status	PASS (EPS=1e-9)

Table 5: Reserve validation metrics for V10.2 Candidate 1.

9 Why the Intermediate Class Matters

The intermediate class matters because it changes the model’s representational topology from a forced two-pole classifier to a three-outcome landscape. Without an explicit intermediate label, borderline trajectories are compelled to collapse into recovery or pathology. That compression causes the model to fight itself across P2, P3, and P4.

A legitimate intermediate class allows liminal trajectories to occupy their own region of state space. This reduces the need for the support gate and crisis shift to compensate for an artifactually binary outcome mapping. The result is a more stable fit to the SPRT target distribution while preserving the other falsification predictions.

10 Formal Model and Code Notes

The Phase 2 dynamical system is implemented in `V10.2_intermediate_dynamics/run_intermediate_dynamics.p`. V10.2 Candidate 1 uses a sigmoid-gated support term and an explicit bounded intermediate-state classifier. The implementation details are summarized in the appendix to keep the main body focused on results.

11 Falsification and Failure Conditions

The current candidate is judged against pre-specified falsification conditions rather than post-hoc fit alone. The key failure conditions are:

- P1 fails if sequence integrity drops below 40% on Profound-or-higher trajectories.
- P2 fails if the support-to-recovery relationship becomes effectively linear ($R^2 > 0.85$).
- P3 fails if the intermediate class becomes larger than either polar cluster.
- P4 fails if crisis pathology is not at least +5pp higher and crisis recovery is not at least -5pp lower.
- Global structure requires revision if JSD exceeds 0.15.

These conditions define the boundaries of the current candidate’s validity and make the release falsifiable in principle.

12 Limitations and Non-Claims

V10.2 Candidate 1 is a simulation-supported candidate, not a completed empirical theory. The results show that the current dynamical structure can satisfy the stated simulation criteria under full confirmation and untouched reserve validation. They do not prove that the same parameters govern human insight, clinical destabilization, mystical experience, or living neural computation.

This release does not claim:

- that the thresholds are universal constants;
- that the model is clinically diagnostic;
- that the model proves a biological-compute substrate claim;
- that the VRP third-substrate claim is empirically complete;
- that simulation fit is equivalent to human validation;
- that all forms of insight, psychosis, creativity, or transformation reduce to this model.

The biological-compute and VRP claims remain candidate extensions. The decisive cross-substrate test is still pending: biological neural cultures must show degradation or routing signatures that are isomorphic to the human and computational substrates under the model’s D/S/A structure.

13 Release Status

V10.2 Candidate 1 is currently a simulation-supported candidate release. It has passed the internal full confirmation run and untouched reserve validation, and it is frozen as `V10.2.candidate.1`. The release is not yet empirically validated or cross-substrate tested.

14 Future Work

The next steps are focused on validation rather than tuning:

- Collect independent human behavioral or neurophysiological data to test the model’s support-threshold and intermediate-state predictions.
- Design cross-substrate biological-compute experiments to evaluate the VRP third-substrate claim.
- Analyze the sensitivity of the intermediate-class boundaries to empirical measurement error.
- Document the conditions under which the current candidate fails, especially in the presence of different support and crisis initialization regimes.

15 Conclusion

V10.2 Candidate 1 is the strongest current simulation-supported release of the Dynamical Model of Realization. It resolves the zero-intermediate weakness of earlier runs, lowers JSD below the good-fit threshold, preserves the P1 stage-sequence prediction, restores strong P2 support nonlinearity, and maintains the P4 crisis-cohort shift in both confirmation and reserve validation.

The main conceptual advance is not merely a better parameter set. It is the recognition that liminal trajectories require their own explicit state class. Once the model stops forcing borderline cases into recovery or pathology, the system no longer has to trade distributional fit against support-threshold nonlinearity and crisis-shift robustness.

This release should therefore be treated as a frozen simulation candidate ready for documentation, review, and empirical follow-up. The next decisive step is no longer tuning. It is validation against independent human behavioral data and cross-substrate biological-compute tests.

16 Appendix

16.1 Release artifacts

The final artifact set includes:

- V10.2_intermediate_dynamics/results/v10.2_candidate_1_metrics.json
- V10.2_intermediate_dynamics/results/v10.2_candidate_1_report.md
- V10.2_intermediate_dynamics/results/v10.2_candidate_1_reserve.csv
- V10.2_intermediate_dynamics/results/v10.2_candidate_1_reserve_summary.md
- V10.2_intermediate_dynamics/V10.2_candidate_1_white_paper.tex

16.2 Implementation details

The final candidate declaration and explicit intermediate-class override logic are shown below.

Listing 1: Final V10.2 Candidate 1 declaration.

```
1 V10_2_CANDIDATE_1 = {
2     'name': 'rescue_g154_g2100_k16_x048',
3     'params': {
4         **PARAMS,
5         'gamma1': 0.54,
6         'gamma2': 1.00,
7         'support_k': 16,
8         'support_x0': 0.48,
9     },
10    'classifier_kwargs': {
11        'D_peak_min': 0.40,
12        'D_final_low': 0.25,
13        'D_final_high': 0.75,
14        'S_low': 0.35,
15        'S_high': 0.68,
16        'A_low': 0.25,
17        'A_high': 0.48,
```

```

18     },
19 }

```

Listing 2: Explicit intermediate classification override.

```

1 def classify_outcome_v10_2(...):
2     ...
3     intermediate_override = (
4         D_peak > D_peak_min
5         and D_final >= D_final_low
6         and D_final <= D_final_high
7         and ((S_final >= S_low and S_final <= S_high)
8              or (A_final >= A_low and A_final <= A_high))
9         and not high_conf_pathology
10        and not high_conf_recovery
11        and R_final < thresholds['R_discard']
12    )
13
14    if intermediate_override:
15        return 7
16    return original

```

16.3 Compilation and Dependencies

This document requires a local $\text{\TeX}$ distribution to compile. The figures are embedded as PNG images generated by the accompanying Python script `generate_v10_2_charts.py`. To compile:

1. Ensure a $\text{\TeX}$ distribution such as $\text{\TeX}$ Live, Mik $\text{\TeX}$, or Mac $\text{\TeX}$ is installed.
2. Place the `figures/` directory beside `V10_2.candidate_1.white_paper.tex`.
3. Run `pdflatex` twice to resolve all cross-references and page breaks:

```

pdflatex V10_2.candidate_1.white_paper.tex
pdflatex V10_2.candidate_1.white_paper.tex

```

The output is `V10_2.candidate_1.white_paper.pdf`. If the figures are missing or the compile fails, check:

- All PNG files exist in the `figures/` subdirectory.
- The working directory is the same as the `.tex` file.
- All required $\text{\LaTeX}$ packages are installed (`pgfplots`, `graphicx`, `hyperref`, etc.).